

CHAPTER 4

CONDITIONAL PROBABILITY, INDEPENDENCE, AND MARKOV CHAINS

4.1 INTRODUCTION

Consider a situation where we want to evaluate the probability $P(A)$ of some event A . Suppose that after finding $P(A)$, we learn that some other event, B , occurred. In many cases, such an information leads to a change in the assessment of the probability of the event A . The symbol used for this new probability will be $P(A|B)$, to be read “conditional probability of A , given B ,” or “probability of event A , given that B occurred.”

■ EXAMPLE 4.1

Conditional probabilities are most easily interpreted as probabilities in subpopulations. Consider an attribute such as color blindness, known to occur much more often among men than among women. If D is the event “a randomly selected person is color blind,” then $P(D)$ refers to the chance of color blindness in the whole population. Suppose now that the person selected is known to be a woman (event W). This information changes the assessment of probability of color blindness to $P(D|W)$, which is now the probability of color blindness in the subpopulation of women.

Questions that might arise here are the following:

1. How to use data on probabilities of color blindness separately among men and women to find the overall chance of color blindness, that is, to find $P(D)$ if we know $P(D|W)$ and $P(D|M)$?
2. How to find the probability that a randomly selected color blind person is a woman, that is, $P(W|D)$?

The first of these questions requires using the weighted average, usually referred to as the formula for total probability (Section 4.3). To answer the second question, one has to use the Bayes' formula (Section 4.4).

The examples and exercises in this chapter are designed to provide practice in recognizing which probabilities are conditional and which are not.

PROBLEMS

4.1.1 A computer file contains data on households in a certain city. Each entry line in this file contains various information about one household: income, socioeconomic status, number of children, their ages, and so on. One data line can then be selected at random (each line has the same probability of being selected). Consequently, probabilities of various events are interpretable as relative frequencies of entries in the data file with the corresponding property.

Let X , Y , and Z be, respectively, the numbers of boys, girls, and cars in the households sampled. Let A be the event that a household has a TV set, and let B be the event that it has a swimming pool.

- (i) Interpret the probabilities below as relative frequencies of occurrence of some attributes in certain subpopulations. (a) $P(A)$. (b) $P(A|Z > 0)$. (c) $P(Z > 0|A)$. (d) $P(X = 0|X + Y = 3)$. (e) $P(B|A^c)$. (f) $P[(X + Y = 0)^c|A \cap B]$. (g) $P(XY = 0|X + Y > 1)$.
- (ii) Use symbols to express probabilities corresponding to the following relative frequencies: (a) Relative frequency of households with two cars. (b) Relative frequency of households with no children among households with at least one car. (c) Relative frequency of households that have both a swimming pool and a TV set, among those who have either a swimming pool or a TV set and have at least one car.

4.2 CONDITIONAL PROBABILITY

The conditional probability of event A given event B is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \text{provided } P(B) > 0. \quad (4.1)$$

The definition of conditional probability in cases when the conditioning event B has probability zero will be discussed in later chapters. In this chapter, it is always assumed, even if not stated explicitly, that the event appearing in the condition has a positive probability.

A motivation of the definition (4.1), based on the frequential interpretation of probability, is the following. As in Chapter 2, let $N(\cdot)$ denote the number of occurrences of the event in parentheses in initial N repetitions of the experiment. Then $P(A|B)$ is to be approximated by the frequency of occurrence of A among those cases where B occurred. Now that B occurred in $N(B)$ cases, and the number of occurrences of A among them is $N(A \cap B)$. Consequently, $P(A|B)$ should be the limit of $N(A \cap B)/N(B) = [N(A \cap B)/N]/[N(B)/N]$, and the latter converges to $P(A \cap B)/P(B)$.

Observe that the conditional probabilities $P(A|B)$, for a fixed event B , satisfy the axioms of probability from Chapter 2. Indeed, nonnegativity holds because $P(A|B)$ is a ratio

of two nonnegative numbers, $P(A \cap B)$ and $P(B)$. Next $P(S|B) = P(S \cap B)/P(B) = P(B)/P(B) = 1$. Finally, if the events $A_1, A_2, \dots$ are mutually exclusive, the same is true for the events $A_1 \cap B, A_2 \cap B, \dots$ so that

$$P\left(\bigcup_k A_k|B\right) = \frac{P(\bigcup_k A_k \cap B)}{P(B)} = \frac{\sum_k P(A_k \cap B)}{P(B)} = \sum_k P(A_k|B).$$

A simple consequence of (4.1) is the formula

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A). \quad (4.2)$$

Notice that the first equality is equivalent to (4.1). The second equality follows from the observation that the left-hand side of (4.2) remains the same when we interchange the roles of A and B . Such an interchange applied to the middle term gives the right-hand side term.

Formula (4.1) shows how to find conditional probability if we have the corresponding unconditional probabilities. Formula (4.2), on the other hand, shows that one may find the probability of an intersection of two events as the product of the conditional probability of one event given the second, and the unconditional probability of the second event. These two ways of using conditional probability will now be illustrated by simple examples.

■ EXAMPLE 4.2

Consider families with two children. Assuming that each of the four combinations of sexes, BB, BG, GB , and GG , is equally likely, find the probability that a family has two boys, given that at least one child is a boy.

SOLUTION. We have

$$P(\text{two boys} \mid \text{at least one boy}) = \frac{P([\text{two boys}] \cap [\text{at least one boy}])}{P(\text{at least one boy})}. \quad (4.3)$$

Since the event “two boys” implies (is contained in) the event “at least one boy,” their intersection is the event “two boys.” Hence, the probability in the numerator equals $1/4$. In the denominator, we have the event $\{BB, BG, GB\}$, and its probability is $3/4$. Thus, the answer is $1/3$. Notice that

$$P(\text{two boys} \mid \text{older child is a boy}) = \frac{P(BB)}{P(BG \text{ or } BB)} = \frac{1}{2}. \quad (4.4)$$

The answers in (4.3) and (4.4) are not the same since the two probabilities refer to families with two boys in different subpopulations. In the first case, we consider all families that have at least one boy, so we eliminate families with two girls. In the second case, we are interested in all families whose older child is a boy, so we again eliminate not only families with two girls but also families with boy being the younger child (combination GB).

■ EXAMPLE 4.3

Assume that in some population, the ratio of the number of men to the number of women is r . Assume also that color blindness occurs among men with frequency p , while among women with frequency p^2 . If you choose a person at random, what is the probability of selecting a woman who is not color blind? (Incidentally, color blindness is a sex-linked attribute, and this is why the frequency of its occurrence among females is the square of the frequency of its occurrence among males; see Example 4.9.)

SOLUTION. Let M, W , and D denote the events “man selected,” “woman selected,” and “selected person is color blind.” Our objective is to find the probability $P(W \cap D^c)$. Using (4.2), we write $P(W \cap D^c) = P(D^c|W)P(W) = P(W|D^c)P(D^c)$. The third term is useless, since the data provide directly neither $P(D^c)$ nor $P(W|D^c)$. Now, using the middle term, we have $P(D^c|W) = 1 - p^2$. To determine $P(W)$, we note that $P(M)/P(W) = [1 - P(W)]/P(W) = r$, which gives the solution $P(W) = 1/(1 + r)$. Consequently, $P(W \cap D^c) = (1 - p^2)/(1 + r)$.

As an immediate consequence of the definition of conditional probability, we have the following theorem:

Theorem 4.2.1 (Chain Rule) *For any events $A_1, A_2, \dots, A_n$, we have*

$$\begin{aligned} P(A_1 \cap A_2 \cap \dots \cap A_n) \\ = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1}), \end{aligned} \quad (4.5)$$

provided $P(A_1 \cap \dots \cap A_{n-1}) > 0$ (which implies that all conditional probabilities appearing in (4.5) are well defined).

Proof: It suffices to write each of the conditional probabilities on the right-hand side as the corresponding ratio of unconditional probabilities according to (4.1). The product cancels then to the probability on the left-hand side. $\square$

It might be helpful to observe that the chain rule is closely related to the counting formula in sampling without replacement in Section 3.2.

■ EXAMPLE 4.4

A man has N keys of which only one opens the door. For some reason, he tries them at random (eliminating the keys that have already been tried). What is the probability that he opens the door on k th attempt?

SOLUTION. Let A_i be the event “ i th attempt unsuccessful.” The problem is then to find $P(A_1 \cap A_2 \cap \dots \cap A_{k-1} \cap A_k^c)$. Applying the chain rule, we obtain

$$\begin{aligned} P(A_1 \cap A_2 \cap \dots \cap A_{k-1} \cap A_k^c) \\ = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P(A_k^c|A_1 \cap \dots \cap A_{k-1}). \end{aligned} \quad (4.6)$$

We have here $P(A_1) = (N - 1)/N$. Next, $P(A_2|A_1) = (N - 2)/(N - 1)$, since after A_1 occurs there are $N - 1$ keys to try, of which $N - 2$ do not open the door. Generally, if $A_1 \cap A_2 \cap \dots \cap A_{j-1}$ occurs, then $j - 1$ keys have been tried and eliminated. This means that there are $N - (j - 1) = N - j + 1$ keys to be tried, of which one fits the door; hence,

$$P(A_j|A_1 \cap A_2 \cap \dots \cap A_{j-1}) = (N - j)/(N - j + 1).$$

Substitution to (4.6) gives the answer $1/N$. It may be surprising that the answer does not depend on k . A success close to the beginning or to the end of the search is as likely as a success closer to the middle. Let us see another solution to the problem. Trying the keys at random without repetition is the same as ordering the keys into a sequence and trying them in this order. The chances that the correct key will appear at the k th

place are the same as for any other place; hence, the probability must be the same for success at any trial.

A convenient way of using the chain rule is related to computing probabilities through *event trees*. This technique is applicable when the outcomes in the sample space $\mathcal{S}$ are sequences of events, typically with the preceding events affecting the probabilities of the subsequent ones. The set of such outcomes can then be depicted as a tree, and probabilities of the outcomes (“branches” of a tree) are calculated according to the chain rule as products of probabilities assigned to consecutive segments of the branch. This general description will be illustrated by an example of an urn scheme, a convenient device for analyzing many real-life problems.

EXAMPLE 4.5

An urn initially contains r red balls and g green balls. A ball is selected at random and returned to the urn. If the chosen ball is red, then a red and b green balls are added. If the chosen ball is green, then c red and d green balls are added. Next, a second ball is drawn, returned, and the process of adding balls is repeated. Finally, the third ball is drawn. Find the probabilities for each possible sequence of colors on three draws: (red, green, green), (red, red, red), and so on.

SOLUTION. Let us use the notation R_1, G_1, R_2, G_2, R_3 , and G_3 for the events that occur at the first, second, and third drawings. The “natural” choice of the sample space $\mathcal{S}$ is

$$\mathcal{S} = \left\{ \begin{array}{l} R_1 \cap R_2 \cap R_3, R_1 \cap R_2 \cap G_3, R_1 \cap G_2 \cap R_3, R_1 \cap G_2 \cap G_3, \\ G_1 \cap R_2 \cap R_3, G_1 \cap R_2 \cap G_3, G_1 \cap G_2 \cap R_3, G_1 \cap G_2 \cap G_3 \end{array} \right\}.$$

We need to find the probabilities such as $P(R_1 \cap G_2 \cap G_3)$. Using the chain rule, we obtain $P(R_1 \cap G_2 \cap G_3) = P(R_1)P(G_2|R_1)P(G_3|R_1 \cap G_2)$. The outcomes are represented as branches of the tree in Figure 4.1, labeled with the corresponding probabilities. By the chain rule, each branch has a probability equal to the product of the probabilities assigned to its segments, for example,

$$P(R_1 \cap G_2 \cap G_3) = \frac{r}{r+g} \times \frac{g+b}{r+g+a+b} \times \frac{g+b+d}{r+g+a+b+c+d}.$$

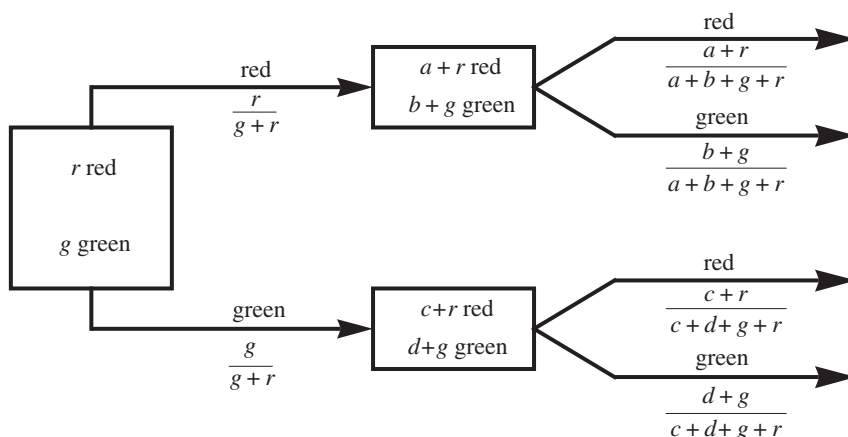


Figure 4.1 Possible results of the two first draws in Example 4.5.

Observe that we do not make any assumptions about the signs of a, b, c , and d , which makes this scheme quite flexible. Therefore, if we want to consider the scheme where the balls sampled are not returned, and no other balls are added, we can put $a = -1, b = c = 0$, and $d = -1$.

PROBLEMS

- 4.2.1** Let events A, B , and C be such that $P(A) > 0, P(B) > 0$, and $P(C) > 0$. Label the following statements as true or false: **(i)** The conditional probability $P(A|B)$ can never exceed the unconditional probability $P(A)$. **(ii)** If $P(A|B) = P(A|C)$, then $P(B) = P(C)$. **(iii)** If A and B are disjoint, then $P(A|A \cup B) = P(A)/[P(A) + P(B)]$. **(iv)** $P(A|B) + P(A^c|B) = 1$. **(v)** $P(A|B) + P(A|B^c) = 1$.
- 4.2.2** Assume that A and B are disjoint. Find $P(A^c|B)$ and $P(A \cup B|A^c)$.
- 4.2.3** Assume that $P(A \cap B) > 0$ and determine $P(A)/[P(A) + P(B)]$ as a function of a , where $a = P(B|A)/P(A|B)$.
- 4.2.4** Find $P(A \cap B)$ if $P(A) = 3/4, P(B) = 1/2, P(A|B) - P(B|A) = 2/9$.
- 4.2.5** $P(A) = P(B) = 1/2, P(A|B) = 1/3$. Find $P(A \cap B^c)$.
- 4.2.6** Find $P(A|B)$, if $P(A^c) = 2P(A)$ and $P(B|A^c) = 2P(B|A) = 0.2$.
- 4.2.7** Find $P[A \cap B \cap C | (A \cap B) \cup (A \cap C)]$ if $P(A) = 0.8, P(B) = 0.4, P(C) = 0.4, P(A \cup B) = 1, P(A \cup C) = 0.9$, and $P(B \cup C) = 0.6$.
- 4.2.8** Events A, B , and C are such that at least one of them occurs. Moreover, $P(A|B) = P(B|C) = P(C|A) = 1/2, P(B \cap C) = 2P(A \cap B) = 4P(C \cap A)/3 = 2P(A \cap B \cap C)$. Find the probability that: **(i)** Exactly one of events A, B , and C occurs. **(ii)** Only B occurs.
- 4.2.9** Three distinct integers are chosen at random from the set $\{1, 2, \dots, 15\}$. Find the probability that: **(i)** Their sum is odd. **(ii)** Their product is odd. **(iii)** The sum is odd if it is known that product is odd. **(iv)** The product is odd if it is known that the sum is odd. **(v)** Answer questions (i)–(iv) if three integers are selected from $\{1, \dots, 20\}$. **(vi)** Generalize answers to (i)–(iv) in the case of selection from $\{1, \dots, n\}$. **(vii)** Answer questions (i)–(iv) by drawing from $\{1, \dots, n\}$, and passing to the limit with $n \rightarrow \infty$.
- 4.2.10** A deck of eight cards contains four jacks and four queens. A pair of cards is drawn at random. Find the probability that both cards are jacks if: **(i)** At least one of the cards is a jack. **(ii)** At least one of the cards is a red jack. **(iii)** One of the cards is a jack of hearts.
- 4.2.11** A fair coin is tossed until a head appears. Given that the first head appeared on an even-numbered toss, what is the probability that it appeared on the $2n$ th toss?
- 4.2.12** A tennis player has the right to two attempts at a serve: If he misses his first serve, he can try again. A serve can be played “fast” or “slow.” If a serve is played fast, the probability that it is good (the ball hits opponent’s court) is A ; the same probability for a slow serve is B . Assume that $0 < A < B$; that is, the fast serve is more difficult (but not impossible) to make.
- If a serve is good, the ball is played, and eventually one of the players wins a point. Let a be the probability that the server wins the point if the serve is fast (and good),

and let b be the same probability for a slow serve. Assume that $0 < b < a$; that is, a fast serve gives a certain advantage to the server (the ball is harder to return, etc.).

The server has four possible strategies: FF , FS , SF , and SS , where FF is “play first serve fast; if missed, play second serve fast,” FS is “play first serve fast; if missed play second serve slow,” and so on. **(i)** Determine the probabilities P_{FF} , P_{FS} , P_{SF} , and P_{SS} of winning the point by the server under the four strategies. **(ii)** Show that the strategy SF is always inferior to the strategy FS . (*Hint*: Consider the difference $P_{FS} - P_{SF}$.) **(iii)** Show that if $Aa > Bb$, then strategy FF is better than FS . **(iv)** Show, by choosing appropriate numbers A , B , a , and b (with $A < B$ and $a > b$), that each of the strategies FF and SS may be best among the four. Explain why such cases do not occur among top players, for whom the best strategy is FS .

4.2.13 Three cards are drawn without replacement from an ordinary deck of cards. Find the probability that: **(i)** The first heart occurs on the third draw. **(ii)** There will be more red than black cards drawn. **(iii)** No two consecutive cards will be of the same value.

4.2.14 An urn contains three red and two green balls. If a red ball is drawn, it is not returned, and one green ball is added to the urn. If a green ball is drawn, it is returned, and two blue balls are added. If a blue ball is drawn, it is simply returned to the urn. Find the probability that in three consecutive draws from the urn, there will be exactly one blue ball drawn.

4.3 PARTITIONS; TOTAL PROBABILITY FORMULA

In this section, we introduce some formulas, important and useful in computing certain probabilities. We begin with the concept of *partition* (used already in Section 1.4).

Definition 4.3.1 We say that a class of events $\mathcal{H} = \{H_1, H_2, \dots\}$ forms a *partition* (of the sample space $\mathcal{S}$) if these events exclude one another and one of them must occur. Formally, this means that

$$H_i \cap H_j = \emptyset \quad \text{for all } i \neq j \quad (4.7)$$

and

$$H_1 \cup H_2 \cup \dots = \mathcal{S}. \quad (4.8)$$

□

We say that a partition is *finite* or *countable*, depending on whether it contains a finite or a countably infinite number of events. In later chapters, we also consider uncountable partitions.

Definition 4.3.2 Given two partitions

$$\mathcal{H} = \{H_1, H_2, \dots\} \quad \text{and} \quad \mathcal{K} = \{K_1, K_2, \dots\},$$

we say that $\mathcal{H}$ is finer than $\mathcal{K}$ ($\mathcal{H}$ is *contained* in $\mathcal{K}$) if for every event H_i in $\mathcal{H}$ there exists an event K_j in $\mathcal{K}$ such that $H_i \subset K_j$. □

■ EXAMPLE 4.6

For any event B , $\mathcal{K} = \{B, B^c\}$ is always a partition of $\mathcal{S}$. A pair of events A and B allows us to define a partition $\mathcal{H}$ into four sets: $H_1 = A \cap B$, $H_2 = A \cap B^c$, $H_3 = A^c \cap B$, and $H_4 = A^c \cap B^c$. We have here $\mathcal{H} \subset \mathcal{K}$, as one can easily verify.

Generally, if $\mathcal{H} = \{H_1, H_2, \dots\}$ and $\mathcal{K} = \{K_1, K_2, \dots\}$ are two partitions, then the class of all sets $H_i \cap K_j$ for H_i in $\mathcal{H}$ and K_j in $\mathcal{K}$ is a partition, called the *intersection* of $\mathcal{H}$ and $\mathcal{K}$ and denoted $\mathcal{H} \times \mathcal{K}$. It is finer than both $\mathcal{H}$ and $\mathcal{K}$.

Finer partitions correspond to more detailed (richer) information, whereas larger partitions are used to represent general information.

■ EXAMPLE 4.7

Consider a data file on employees of some company. Let $\mathcal{H} = \{H_0, H_1, \dots\}$ be the partition of data by level of education, and let $\mathcal{K} = \{K_0, K_1, \dots\}$ be the partition of data corresponding to the number of full years of employment in the company. Knowledge of the category in $\mathcal{H}$ to which a data entry belongs gives information about the education level of the employee. Knowledge of the category in the intersection $\mathcal{H} \times \mathcal{K}$ gives information about both education level and number of years of employment, hence gives more information than either of $\mathcal{H}$ or $\mathcal{K}$ alone.

It will be convenient to introduce the following definition:

Definition 4.3.3 A partition $\mathcal{H} = \{H_1, H_2, \dots\}$ is called *positive*, if $P(H_i) > 0$ for all H_i in $\mathcal{H}$. □

We can now formulate the following simple theorem called the *formula for total probability* providing a valuable tool for computing probabilities.

Theorem 4.3.1 Let $\mathcal{H} = \{H_j, j = 1, 2, \dots\}$ be a positive (finite or infinite) partition and let A be an arbitrary event. Then

$$P(A) = P(A|H_1)P(H_1) + P(A|H_2)P(H_2) + \dots + P(A|H_n)P(H_n) \quad (4.9)$$

in the case of a finite partition $\mathcal{H}$, or

$$P(A) = P(A|H_1)P(H_1) + P(A|H_2)P(H_2) + \dots \quad (4.10)$$

in the case of an infinite countable partition $\mathcal{H}$.

Proof: Using (4.7), (4.8), and (4.1), we have

$$\begin{aligned} P(A) &= P((A \cap H_1) \cup (A \cap H_2) \cup \dots) \\ &= P(A \cap H_1) + P(A \cap H_2) + \dots \\ &= P(A|H_1)P(H_1) + P(A|H_2)P(H_2) + \dots \end{aligned}$$

The summation may be finite or not, depending on whether the partition $\mathcal{H}$ is finite. □

One may think of the conditional probability $P(A|H_k)$ as the relative frequency of attribute A in the subpopulation corresponding to event H_k . Then formulas (4.9) and (4.10) give the overall frequency of attribute A in the whole population, expressed as the weighted average, with weights $P(H_k)$, $k = 1, 2, \dots$ being the contributions of the k th subpopulation to the whole population.

In the simplest case of a positive partition into two sets ($\{B, B^c\}$), formula (4.9) reduces to

$$\begin{aligned} P(A) &= P(A|B)P(B) + P(A|B^c)P(B^c) \\ &= P(A|B)P(B) + P(A|B^c)(1 - P(B)). \end{aligned}$$

■ EXAMPLE 4.8

Recall Example 4.3. In a certain group of people the ratio of the number of men to the number of women is r . It is known that the incidence of color blindness among men is p , and the incidence of color blindness among women is p^2 . What is the probability that a person randomly chosen from this group is color blind?

SOLUTION. We use the formula for total probability, with events M and W (choice of man and choice of woman) as a partition. The probability of the event D (the person selected is color blind) is then $P(D) = P(D|M)P(M) + P(D|W)P(W)$. Since $P(M) = r/(1+r)$ and $P(W) = 1/(1+r)$ (see Example 4.3), the answer is $P(D) = pr/(1+r) + p^2/(1+r) = p(p+r)/(1+r)$.

■ EXAMPLE 4.9

The reason why the frequency of color blindness among females is the square of the corresponding frequency among males is that the gene responsible for color blindness (as well as the gene responsible for other sex-linked attributes, e.g., hemophilia) is located on the X chromosome. There are two sex chromosomes, X and Y , and every person has a pair of such chromosomes, with individuals of the type XX being females and XY being males.

The color blindness gene has two forms (alleles), say C and c , with form c being recessive and causing color blindness. Because every man has one X chromosome, he therefore is either of category C or c , the latter being color-blind. Women, having a pair of X chromosomes, are of one of three categories: (1) CC , which we will call “normal;” (2) Cc , the so-called carriers (i.e., persons who are not color-blind but are capable of transmitting the color blindness gene c to their offspring), and (3) cc , women who are color-blind.

Let p be the relative frequency of the form c in the population of genes in question. Then p is also the relative frequency of color-blind men (since each man carries one color-blindness gene, either C or c). Now, let u and v denote the relative frequency of carriers and of color-blind women, respectively. To find u and v , we proceed as follows: So far we know that a woman is color-blind if her father is color-blind and either (1) her mother is color-blind, or (2) her mother is a carrier and transmits the gene c to the daughter. This gives, using the formula for total probability, the relation

$$v = p \left(v + \frac{1}{2}u \right). \quad (4.11)$$

On the other hand, a woman is a carrier if either (1) her father is color-blind, and her mother is “normal,” (2) her father is color-blind, and her mother is a carrier who transmits to her the gene C , (3) her father is “normal,” and her mother is a carrier who transmits gene c , or (4) her father is “normal,” and her mother is color-blind.

This gives the relation

$$u = p \left[(1 - u - v) + \frac{1}{2}u \right] + (1 - p) \left[\frac{1}{2}u + v \right]. \quad (4.12)$$

The solution of the pair of equations (4.11) and (4.12) gives $v = p^2$, $u = 2p(1 - p)$. The relative frequency of women who are “normal” (i.e., neither color-blind nor a carrier) is $1 - u - v = (1 - p)^2$, while the relative frequency of color-blind women is p^2 , as asserted.

The formula for total probability can be extended to conditional probabilities as follows:

Theorem 4.3.2 *Let $\mathcal{H} = \{H_1, H_2, \dots\}$ be a positive partition of the sample space, and let A, B be two events with $P(B) > 0$. Then*

$$P(A|B) = \sum_j P(A|B \cap H_j)P(H_j|B). \quad (4.13)$$

Proof: The right-hand side can be written as⁹

$$\begin{aligned} \sum_j \frac{P(A \cap B \cap H_j)}{P(B \cap H_j)} \times \frac{P(B \cap H_j)}{P(B)} &= \frac{1}{P(B)} \sum_j P(A \cap B \cap H_j) \\ &= \frac{1}{P(B)} P\left((A \cap B) \cap \bigcup_j H_j\right) = \frac{P(A \cap B)}{P(B)} = P(A|B). \end{aligned}$$

□

One can also consider “coarsening” of a partition $\mathcal{H}'$ by grouping its sets, thus forming a new partition $\mathcal{H}$ corresponding to a less precise information.

Theorem 4.3.3 *Let $\mathcal{H}, \mathcal{H}'$ be two partitions, with $\mathcal{H}$ being a coarsening of $\mathcal{H}'$. Then for any event A ,*

$$P_{\mathcal{H}}(A|H_i) = \sum_{j=1}^k P_{\mathcal{H}'}(A|H'_{ij})P(H'_{ij}|H_i),$$

where $H_i = H'_{i_1} \cup \dots \cup H'_{i_k}$, $H_i \in \mathcal{H}$, and $H'_{ij} \in \mathcal{H}'$.

Thus, the conditional probability with respect to a coarser partition is the weighted average of probabilities with respect to the finer partition.

Proof: The proof is obtained by simple algebra. If $\{H'_{ij}, \dots, H'_{i_k}\}$ is a partition of H_i , then

$$\begin{aligned} P(A|H_i) &= P(A \cap H_i|H_i) = P\left(A \cap \bigcup_{j=1}^k H'_{ij}|H_i\right) \\ &= \sum_{j=1}^k P(A \cap H'_{ij}|H_i) = \sum_{j=1}^k \frac{P(A \cap H'_{ij} \cap H_i)}{P(H_i)} \\ &= \sum_{j=1}^k \frac{P(A \cap H'_{ij})}{P(H_i)} = \sum_{j=1}^k \frac{P(A|H'_{ij})P(H'_{ij})}{P(H_i)} \\ &= \sum_{j=1}^k P(A|H'_{ij}) \frac{P(H'_{ij} \cap H_i)}{P(H_i)} = \sum_{j=1}^k P(A|H'_{ij})P(H'_{ij}|H_i). \end{aligned}$$

□

⁹Formally, the summation in (4.13) extends only over those j for which $P(B \cap H_j) > 0$, otherwise, $P(A|B)$ is not defined. It does not matter here since $P(B \cap H_j) = 0$ implies $P(H_j|B) = 0$.

PROBLEMS

- 4.3.1** An event W occurs with probability 0.4. If A occurs, then the probability of W is 0.6; if A does not occur but B occurs, the probability of W is 0.1. However, if neither A nor B occurs, the probability of W is 0.5. Finally, if A does not occur, the probability of B is 0.3. Find $P(A)$.
- 4.3.2** Suppose that initially the urn contains one red and two green balls. We draw a ball and return it to the urn, adding three red, one green, and two blue balls if a red ball was drawn, and three green and one blue ball if a green ball was drawn. Then a ball is drawn from the urn. Find the probability that both selected balls are: **(i)** Blue. **(ii)** Red. **(iii)** Green. **(iv)** Of the same color.
- 4.3.3** Suppose that in Problem 4.3.2 we return the second ball to the urn, and add new balls as described, with the condition that if the second ball is blue, we add one ball of each color. Then we draw the third ball. What is the probability that the third ball is: **(i)** Blue? **(ii)** Blue if the first ball was red? **(iii)** Blue if the second ball was red? **(iv)** Blue if no blue ball was drawn on any of the preceding draws?
- 4.3.4** Let A and B be two events with $P(B) > 0$, and let $C_1, C_2, \dots$ be a possible partition of a sample space. Prove or disprove the following formulas:

$$P(A|B) = \sum_i P(A|B \cap C_i)P(C_i),$$

$$P(A|B) = \sum_i P(A|B \cap C_i)P(B|C_i)P(C_i).$$

- 4.3.5 (Tom Sawyer Problem)** You are given a task, say painting a fence. The probability that the task will be completed if k friends are helping you is p_k ($k = 0, 1, \dots$). If j friends already helped you, the probability that the $(j+1)$ st will also help is π_j ($j = 0, 1, \dots$). On the other hand, if the j th friend did not help, then the $(j+1)$ st will not help either. **(i)** Find the probability that the task will be completed. **(ii)** Solve part (i) if $p_k = 1 - \lambda^k$, $\pi_j = \mu$ for all j .
- 4.3.6** Recall Example 4.9. Find the probability that the mother is a carrier if: **(i)** Both father and son are color-blind, and the mother is not. **(ii)** It is known only that the son is color-blind. **(iii)** The son is color-blind, but the parents are not.

4.4 BAYES' FORMULA

Let us consider now a following question: If it is known that a certain event A occurred, and its conditional probabilities $P(A|H_1), P(A|H_2), \dots$ for partition $H_1, H_2, \dots$ are known, can we determine the probability of the event H_k of the partition? The answer is contained in the following time-honored theorem, dating back to the eighteenth century.

Theorem 4.4.1 (Bayes' Formula) Let $\mathcal{H} = \{H_1, H_2, \dots\}$ be a positive partition of $\mathcal{S}$, and A be an event with $P(A) > 0$. Then for any event H_k of the partition $\mathcal{H}$,

$$P(H_k|A) = \frac{P(A|H_k)P(H_k)}{P(A|H_1)P(H_1) + \dots + P(A|H_n)P(H_n)}$$

in the case of a finite partition $\mathcal{H}$, and

$$P(H_k|A) = \frac{P(A|H_k)P(H_k)}{P(A|H_1)P(H_1) + P(A|H_2)P(H_2) + \cdots}$$

when partition $\mathcal{H}$ is countably infinite.

Proof: Using formula (4.1) twice,

$$P(H_k|A) = \frac{P(H_k \cap A)}{P(A)} = \frac{P(A|H_k)P(H_k)}{P(A)}.$$

The theorem follows if one replaces the denominator on the right-hand side by the formulas (4.9) and (4.10). $\square$

In the case of a partition (positive) into two events, $\mathcal{H} = \{B, B^c\}$, and any event A with $P(A) > 0$, we have

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}.$$

One of the interpretations of Bayes' theorem is when the partition $\mathcal{H}$ represents all possible mutually exclusive conditions (states of nature, hypotheses, etc.) that are logically possible. The probabilities $P(H_1), P(H_2), \dots$ represent the prior knowledge, experience, or belief about the likelihood of $H_1, H_2, \dots$. An event A is then observed, and this fact usually modifies the probabilities of H_i 's. Accordingly, $P(H_i)$ and $P(H_i|A)$ are often called *prior* and *posterior* probabilities of H_i .

■ EXAMPLE 4.10

Returning to Example 4.8, suppose that the person that we randomly selected is color-blind. Intuitively, this should increase the chance that this person is a man. Indeed, we have here

$$P(M|D) = \frac{P(D|M)P(M)}{P(D|M)P(M) + P(D|W)P(W)} = \frac{\frac{pr}{1+r}}{\frac{pr}{1+r} + \frac{p^2}{1+r}} = \frac{r}{r+p},$$

a quantity close to 1 if p is small.

As mentioned above, Bayes' theorem can be used to reassess the probabilities of "states of nature" or "hypotheses" based on the additional evidence. How should one reassess the probabilities when the evidence comes sequentially? Specifically, if the evidence comes in two portions, say A' followed by A'' , should one modify the prior probabilities given $A' \cap A''$, or should one first obtain posterior probabilities given A' and then use these posteriors as new priors, to be modified given A'' ?

The answer, as might be expected, is that it does not matter. We have the following theorem:

Theorem 4.4.2 (Updating the Evidence) *Let $\mathcal{H} = \{H_1, H_2, \dots\}$ be a partition, and let A' and A'' be two events. If $P(A' \cap A'') > 0$, then for every event H_k in partition $\mathcal{H}$, we have*

$$\begin{aligned} P(H_k|A' \cap A'') &= \frac{P(A' \cap A''|H_k)P(H_k)}{\sum P(A' \cap A''|H_j)P(H_j)} \\ &= \frac{P(A''|H_k \cap A')P(H_k|A')}{\sum P(A''|H_j \cap A')P(H_j|A')}. \end{aligned} \tag{4.14}$$

Proof: The middle term is Bayes' formula applied to the left-hand side. We write

$$\begin{aligned} P(A' \cap A'' | H_i) P(H_i) &= P(A' \cap A'' \cap H_i) \\ &= P(A'' | A' \cap H_i) P(A' \cap H_i) \\ &= P(A'' | H_i \cap A') P(H_i | A') P(A') \end{aligned}$$

to show the equality of the middle and right-hand-side terms. $\square$

■ EXAMPLE 4.11

An urn contains two coins: One is a regular coin, with heads and tails, while the other has heads on both sides. One coin is chosen at random from the urn and tossed n times. The results are all heads. What is the probability that the coin tossed is a two-headed one?

SOLUTION. Intuitively, for a large n we expect the probability that the coin selected has two heads to be close to 1, since it is increasingly unlikely to get n heads in a row with a regular coin. Let H_1 and H_2 be the events “regular coin was chosen” and “coin with two heads was chosen,” respectively. Clearly, H_1 and H_2 form a partition. Let the prior probabilities be $P(H_1) = P(H_2) = 1/2$, and let A_n be the event “ n heads in a row”; our objective is to find $P(H_2 | A_n)$. Since $P(A_n | H_2) = 1$ for all n (this coin will only give heads), and $P(A_n | H_1) = 1/2^n$, by Bayes' theorem we have

$$P(H_2 | A_n) = \frac{P(A_n | H_2) P(H_2)}{P(A_n | H_1) P(H_1) + P(A_n | H_2) P(H_2)} = \frac{2^n}{2^n + 1}. \quad (4.15)$$

As expected, the probability (4.15) does approach 1 as n increases.

Suppose now that after A_n was observed, an additional m tosses again produced only heads (event B_m). Because $A_n \cap B_m$ is the same as A_{n+m} , the posterior probability of the two-headed coin (H_2) given A_{n+m} is $2^{n+m}/(2^{n+m} + 1)$, after we replace $n + m$ for n in (4.15). Using the second part of formula (4.14), and the fact that $P(B_m | H_i \cap A_n) = P(B_m | H_i)$, $i = 1, 2$, we obtain

$$\begin{aligned} P(H_2 | A_n \cap B_m) &= \frac{P(B_m | H_2) P(H_2 | A_n)}{P(B_m | H_1) P(H_1 | A_n) + P(B_m | H_2) P(H_2 | A_n)} \\ &= \frac{1 \times \frac{2^n}{2^n + 1}}{\frac{1}{2^m} \times \frac{1}{2^n + 1} + 1 \times \frac{2^n}{2^n + 1}} = \frac{2^{n+m}}{2^{n+m} + 1}, \end{aligned}$$

which agrees with the result of updating “all at once.”

It might seem at first that this problem is purely artificial, invented for the sole purpose of providing practice for students. This is not so; the problem is of importance in breeding and selection. Consider a gene with two forms (alleles) A and a . Assume that A is dominant and a is recessive so that we have two phenotypes: Individuals aa can be distinguished from the others, but individuals AA and Aa cannot be told apart. Moreover, assume that allele a is undesirable and we want to eliminate it, ultimately producing a pure strain of individuals of type AA only. The problem then lies in eliminating individuals Aa . One of the ways to do it is to allow cross-breeding between a tested individual (of type AA or Aa) with an individual of type aa (you may think here of plants that can easily be cross-pollinated). If the tested individual is of the type AA (“coin with two heads”), all offsprings will be of type Aa .

However, if the tested individual is of the type Aa (“regular coin”), about half of the offspring will be of type Aa and half of type aa (corresponding to results of tosses being heads or tails with probability $1/2$). Now, if n offspring are of type Aa , the posterior probability that the tested individual is AA can be computed as above, except that the prior probability need not be $1/2$ (it may be assessed from the genetic theory and other information about the tested individual). Usually, the breeder will accept an individual as a pure strain AA if the posterior probability that it is AA exceeds a threshold such as 0.99 or so, hence, if sufficiently many offspring of type Aa have been observed.

■ EXAMPLE 4.12

In the famous Monty Hall problem, there are three curtains, say A , B , and C , of which two hide nothing while behind the third there is a Big Prize. The Big Prize is won if it is guessed correctly which curtain hides it. You choose one of the curtains, say A . Before curtain A is pulled to reveal what is behind it, the game host pulls one of the two other curtains, say B , and shows that there is nothing behind it. He then offers you the option to change your decision (from curtain A to curtain C). Should you stick to your original choice (A) or change to C (or does it matter)? The answer to this question is counterintuitive and has stirred a lot of controversy. The common conviction is that it does not matter: There are two closed curtains, and the chances of the Big Prize being behind either of them must be 50–50, so it is irrelevant whether or not you change your choice.

Actually, the answer is that *you should* switch to C ; this way you double your chance of winning.

We assume that (1) the game host pulls one of the two curtains (B or C) that does not hide the Big Prize (this implies that the host knows where the Big Prize is), and (2) if both curtains B and C have nothing behind them, the host selects one at random.

Let A , B , and C be the events “Big Prize is behind curtain A ” (respectively, B and C). We assume the original choice was curtain A , and let B^* be the event “host shows that there is nothing behind curtain B .” We want $P(A|B^*)$. By Bayes’ formula, taking A , B , C as a partition and assuming $P(A) = P(B) = P(C) = 1/3$,

$$\begin{aligned} P(A|B^*) &= \frac{P(B^*|A)P(A)}{P(B^*|A)P(A) + P(B^*|B)P(B) + P(B^*|C)P(C)} \\ &= \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2} \times \frac{1}{3} + 0 \times \frac{1}{3} + 1 \times \frac{1}{3}} = \frac{1}{3}. \end{aligned}$$

Since $P(B|B^*) = 0$, we must have $P(C|B^*) = 2/3$, so the chances of finding the Big Prize behind curtain C (i.e., winning if one changes the original choice) is $2/3$ and *not* $1/2$.

Here is an argument that might help convince people who claimed originally that the chances were $1/2$. Imagine that John and Peter are going to play the game a large number of times, say 300 times each. Each originally chooses A , and after being given a choice to switch, John never switches while Peter always switches.

John wins only if his original choice was correct, which happens in about 33% times (we assume that the Big Prize is each time randomly located behind one of the curtains A , B , and C). He may expect to win about 100 times out of 300.

On the other hand, Peter wins whenever his original choice was *wrong* (because then switching is a correct choice). Consequently, he wins in about 200 cases out of 300.

PROBLEMS

- 4.4.1** Suppose that medical science has developed a test for a certain disease that is 95% accurate, on both those who do and those who do not have the disease. If the incidence rate of this disease in the population is 5%, find the probability that a person: **(i)** Has the disease when the test is positive. **(ii)** Does not have the disease when the test is negative.
- 4.4.2** Two different suppliers, A and B, provide the manufacturer with the same part. All supplies of this part are kept in a large bin. In the past 2% of all parts supplied by A and 4% of parts supplied by B have been defective. Moreover, A supplies three times as many parts as B. Suppose that you reach into the bin and select a part. **(i)** Find the probability that this part is defective. **(ii)** If the part is nondefective, find the probability that it was supplied by B?
- 4.4.3** An urn originally contains three blue and two green chips. A chip is chosen at random from the urn, returned, and four chips of the opposite color are added to the urn. Then a second chip is drawn. Find the probability that: **(i)** The second chip is blue. **(ii)** Both chips are of the same color. **(iii)** The first chip was green if the second chip is blue.
- 4.4.4** One box contains six red and three green balls. The second box has six red and four green balls. A box is chosen at random. From this box two balls are selected and found to be green. Find the probability that the pair was drawn from the first box if the draws are: **(i)** Without replacement. **(ii)** With replacement.
- 4.4.5** Suppose that box A contains four red and five green chips and box B contains six red and three green chips. A chip is chosen at random from box A and placed in box B. Finally, a chip is chosen at random from those now in box B. What is the probability that a green chip was transferred given that a red chip was drawn from box B?
- 4.4.6** We have three dice, each with numbers $x = 1, \dots, 6$, and with probabilities as follows: die 1: $p(x) = 1/6$, die 2: $p(x) = (7 - x)/21$, die 3: $p(x) = x^2/91$. A die is selected at random, tossed, and the number 4 appears. What is the probability that it is die 2 that was tossed?
- 4.4.7** Players A and B draw balls in turn, without replacement, from an urn containing three red and four green balls. A draws first. The winner is the person who draws the first red ball. Given that A won, what is the probability that A drew a red ball on the first draw?
- 4.4.8** Twenty chips are placed in a box. On one side each chip has either red (10), blue (5), or green (5) color. On the other side each chip has either one or two dots: six red chips have one dot, while four have two dots, exactly two green and three blue chips have two dots. One chip was randomly selected from the box. If it has two dots, what is the probability that it is red on the other side?
- 4.4.9** A prisoner is sentenced to life in prison. One day the warden comes to him and offers to toss a fair coin for either getting free or being put to death. After some deliberation the prisoner refuses, on the ground that it is too much risk: He argues that he may escape, or be pardoned, and so on. The warden asks him if he would agree to play such a game if the odds for death were 1:9 or less. The prisoner agrees.
Here is the game: An urn contains coins, labeled with digits on both sides. There is one coin labeled 1 and 2. There are nine coins labeled 2 and 3, and there are 81 coins

labeled 3 and 4. A coin is to be selected at random by the warden and tossed. The prisoner can see the upper face and has to guess correctly the other face of the coin.

The prisoner decides to use the following guessing strategy: if 1 shows up, he would guess that the other side is 2 (and win). Similarly, if 4 shows up, he would guess that the other side is 3, and win. If 2 shows up, he would guess that the other side is 3, since there is one coin with 1 against nine coins with 3 on the other side. Similarly, if 3 shows up, he would guess 4, since there are nine coins with 2, against 81 coins with 4.

A coin was chosen, tossed, and 2 appeared. The prisoner was about to say “3,” when the following doubts occurred to him: “Suppose that this coin has indeed 3 on the other side. It is therefore a 2–3 coin. But this coin, before it was tossed, had equal chances of showing 2 and showing 3. But if it had shown 3, I would have guessed 4, and be put to death. So I played a game with fifty–fifty chances for death, which I decided not to play at the start.”

As the story goes, the prisoner decided not to play. So he spent the rest of his life in prison contemplating whether or not he did the right thing. Find the probability that the prisoner would go free computed before a coin is selected.

4.5 INDEPENDENCE

The notion of the conditional probability $P(A|B)$ introduced in Section 4.4 concerned the modification of the probability of an event A in light of the information that some other event B has occurred. Obviously, an important special case here when is such information is irrelevant: Whether or not B has occurred, the chances of A remain the same. In such a case, we say that the event A is *independent* of the event B . As we will see, the relation of independence defined in this way is symmetric: when A is independent of B , then B is also independent of A .

The essence of the idea above is that A is independent of B whenever $P(A|B) = P(A)$. Using (4.1), this implies that $P(A \cap B)/P(B) = P(A)$, and multiplying by $P(B)$ —which we may do, since $P(B) > 0$ by assumption—we obtain $P(A \cap B) = P(A)P(B)$. This relation is symmetric in A and in B , as asserted; moreover, it holds also if one or both events have probability zero.

Consequently, we introduce the following definition:

Definition 4.5.1 We say that two events, A and B , are *independent* if their probabilities satisfy the *multiplication rule*:

$$P(A \cap B) = P(A)P(B). \quad (4.16)$$

□

In practice, (4.16) is used in two ways. First, we can compute both sides separately and compare them to check whether or not the two events are independent. More often, however, we *assume* that A and B are independent and use (4.16) for determining the probability of their intersection (joint occurrence). Typically, the assumption of independence is justified on intuitive grounds (e.g., when the events A and B do not influence each other).

■ EXAMPLE 4.13

A box contains r red and g green balls. We draw a ball at random from the box, and then we draw another ball. Let R_1 and R_2 be the events “red ball on the first (second) draw.” The question now is whether the events R_1 and R_2 are independent.

SOLUTION. The answer depends crucially on what happens after the first draw: Is the first ball returned to the box before the second draw or not? More generally, is the content of the box modified in any way by the first draw?

Suppose first that the ball drawn is returned to the box (the scheme known under the name *sampling with replacement*). Then $P(R_1) = P(R_2) = r/(r+g)$; hence, $P(R_1)P(R_2) = [r/(r+g)]^2$. To check the independence, we have to compute $P(R_1 \cap R_2)$. Recall that the two draws with replacement may produce $(r+g)^2$ distinct results, by the product rule of the preceding chapter. The number of ways one can draw two red balls is—again, by the product rule—equal to r^2 . Consequently, $P(R_1 \cap R_2) = r^2/(r+g)^2$, which is the same as $P(R_1)P(R_2)$. This shows that events R_1 and R_2 are independent.

If we do not return the ball to the box (*sampling without replacement*), we can proceed as follows: We have $P(R_1 \cap R_2) = P(R_1)P(R_2|R_1)$ by (4.2), and $P(R_2|R_1) = (r-1)/(r+g-1)$, since if R_1 occurs, there remain $r+g-1$ balls in the box, of which $r-1$ are red. Consequently, $P(R_1 \cap R_2) = r(r-1)/(r+g)(r+g-1)$. To verify whether or not this last quantity equals $P(R_1)P(R_2)$, we need to evaluate $P(R_2)$. Observe that the probability of a red ball on the second draw is a random quantity depending on the outcome of the first draw. Taking events R_1 and G_1 as a partition and using the total probability formula, we can write $P(R_2) = P(R_2|R_1)P(R_1) + P(R_2|G_1)P(G_1)$, which equals $r/(r+g)$, so that $P(R_1 \cap R_2)$ is not the same as $P(R_1)P(R_2)$. Therefore, events R_1 and R_2 are dependent.

We will prove next an important property of independent events that will facilitate many calculations.

Theorem 4.5.1 *If the events A and B are independent, so are the events A and B^c , A^c , and B , as well as A^c and B^c .*

Proof: It is sufficient to prove only the independence of A and B^c . Indeed, the independence of the second pair will follow by symmetry, and the independence of the third pair will follow by successive application of the two already proved statements.

Thus, assume that $P(A \cap B) = P(A)P(B)$ and consider the event $A \cap B^c$. Since $A = (A \cap B) \cup (A \cap B^c)$, we have $P(A) = P(A \cap B) + P(A \cap B^c)$ so that

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A \cap B) = P(A) - P(A)P(B) \\ &= P(A)(1 - P(B)) = P(A)P(B^c), \end{aligned}$$

which had to be shown. □

As shown in Example 4.13, the difference between sampling with and without replacement lies in the fact that the first leads to independent outcomes, while the second does not. However, when the population is large, this difference is negligible.

Specifically, we may regard two events A and B as “almost independent” if the difference between $P(A \cap B)$ and $P(A)P(B)$ is small. Let us compute the difference

$$Q = |P(R_1 \cap R_2) - P(R_1)P(R_2)|$$

in the case of sampling without replacement (obviously, for sampling with replacement, this difference is zero). We have here

$$Q = \left| \left(\frac{r}{r+g} \right) \left[\left(\frac{r-1}{r+g-1} \right) - \left(\frac{r}{r+g} \right) \right] \right| = \frac{rg}{(r+g)^2(r+g-1)},$$

a quantity that is small when either r or g is large (or both are).

It appears that we might take $|P(A \cap B) - P(A)P(B)|$ as a measure of degree of dependence between events A and B . This is indeed the case, except that we have to take into account the fact that this difference may be small simply because one of the events A or B has small probability. Consequently, we need to “standardize” the difference. This leads to the following definition:

Definition 4.5.2 Assume that $0 < P(A) < 1$ and $0 < P(B) < 1$. The quantity

$$r(A, B) = \frac{P(A \cap B) - P(A)P(B)}{\sqrt{P(A)(1 - P(A))P(B)(1 - P(B))}} \quad (4.17)$$

will be called the *coefficient of correlation* between events A and B . $\square$

The coefficient of correlation is a measure of dependence in the sense specified by the following theorem:

Theorem 4.5.2 *The coefficient of correlation $r(A, B)$ is zero if and only if A and B are independent. Moreover, we have $|r(A, B)| \leq 1$, with $r(A, B) = 1$ if and only if $A = B$, and $r(A, B) = -1$ if and only if $A = B^c$.*

Proof: The proof is by elementary checking, except for the “only if” parts of the last two statements. These parts follow from the general theorem on the properties of correlation coefficients for random variables, to be discussed in Chapter 6. $\square$

We can extend the concept of independence to the case of more than two events.

Definition 4.5.3 We say that the events $A_1, A_2, \dots, A_n$ are *independent* if for any set of indices $i_1, i_2, \dots, i_k$ with $1 \leq i_1 < i_2 < \dots < i_k \leq n$, we have the following multiplication rule:

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = P(A_{i_1})P(A_{i_2}) \dots P(A_{i_k}). \quad (4.18)$$

The events in an infinite sequence $A_1, A_2, \dots$ are called independent if for every n , the first n events of the sequence are independent. $\square$

Thus, in the case of n events, we have one condition for every subset of the size $k \geq 2$, that is, for $k = 1$, condition (4.18) is satisfied. Since the number of all subsets of a set with n elements is 2^n , the number of conditions in (4.18) is $2^n - n - 1$.

In the case of three events, A, B, C , the definition (4.18) represents four conditions: $P(A \cap B \cap C) = P(A)P(B)P(C)$, and three conditions for pairs, $P(A \cap B) = P(A)P(B)$, $P(A \cap C) = P(A)P(C)$, and $P(B \cap C) = P(B)P(C)$. In case of four events A, B, C , and D , the multiplication rule (4.18) must hold for the quadruplet (A, B, C, D) , for four triplets (A, B, C) , (A, B, D) , (A, C, D) , and (B, C, D) , and for six pairs (A, B) , (A, C) , (A, D) , (B, C) , (B, D) , and (C, D) .

The question arises: Do we really need so many conditions? Taking the simplest case of $n = 3$ events, it might seem that independence of all three possible pairs, namely (A, B) , (A, C) , and (B, C) , should imply the condition $P(A \cap B \cap C) = P(A)P(B)P(C)$, a direct analogue of the defining condition for pairs of events.

Conversely, one might expect that the multiplication rule $P(A \cap B \cap C) = P(A)P(B)P(C)$ implies independence of pairs (A, B) , (A, C) , and (B, C) , as well as conditions of the kind $P(A \cap B^c \cap C^c) = P(A)P(B^c)P(C^c)$, and so on. In fact, none of these implications is true, as will be shown by examples.

■ **EXAMPLE 4.14**

Independence in pairs does not imply the multiplication rule for more events. We will use the case of three events A, B , and C , and show that the condition $P(A \cap B \cap C) = P(A)P(B)P(C)$ need not hold, even if all three pairs of events are independent.

Suppose that we toss a die twice and let A = “odd outcome on the first toss,” B = “odd outcome on the second toss,” and C = “sum odd.” We have $P(A) = P(B) = P(C) = 1/2$. By simple counting, we obtain $P(A \cap B) = P(A \cap C) = P(B \cap C) = 1/4$, so each of the possible pairs is independent. However, $P(A)P(B)P(C) = 1/8$, while $P(A \cap B \cap C) = 0$, since the events A, B , and C are mutually exclusive. If A and B hold, then the sum of outcomes must be even so that C does not occur.

■ **EXAMPLE 4.15**

Multiplication rule does not imply pairwise independence. We will now show that independence in pairs is not implied by the condition

$$P(A \cap B \cap C) = P(A)P(B)P(C).$$

We take two events A and B that are dependent, and then take C with $P(C) = 0$. Let the sample space be $S = \{a, b, c, d, \dots\}$, and let $A = \{a, d\}$, $B = \{b, d\}$, and $C = \{c, d\}$. Moreover, let $P(a) = P(b) = P(c) = p - p^3$, and $P(d) = p^3$, where $p > 0$ satisfies the inequality $3(p - p^3) + p^3 \leq 1$. The remaining points (if any) have arbitrary probabilities, subject to the usual constraint that the sum of probabilities is 1. We have here $P(A) = P(B) = P(C) = (p - p^3) + p^3 = p$; hence, $P(A \cap B \cap C) = P(A)P(B)P(C) = p^3$. However, since $A \cap B = A \cap C = B \cap C = A \cap B \cap C$, we have $P(A \cap B) = p^3$ while $P(A)P(B) = p^2$, and similarly for other pairs.

The possibility of events being pairwise independent, but not totally (mutually) independent, is mainly of theoretical interest. Unless explicitly stated otherwise, whenever we speak of independent events, we will always mean mutual independence in the sense of Definition 4.5.3.

Let us state here the following analogue of Theorem 4.5.1 for the case of n events.

Theorem 4.5.3 *If the events $A_1, A_2, \dots, A_n$ are independent, the same is true for events $A'_1, A'_2, \dots, A'_n$, where for each k the event A'_k stands for either A_k or its complement A_k^c .*

The following theorem provides one of the most commonly used “tricks of the trade” in probability theory:

Theorem 4.5.4 *If the events $A_1, A_2, \dots, A_n$ are independent, then*

$$P(A_1 \cup \dots \cup A_n) = 1 - [1 - P(A_1)][1 - P(A_2)] \cdots [1 - P(A_n)].$$

Proof: The proof uses the fact that can be summarized as $P(\text{at least one}) = 1 - P(\text{none})$. By De Morgan’s law, $(A_1 \cup A_2 \cup \dots \cup A_n)^c = A_1^c \cap A_2^c \cap \dots \cap A_n^c$. Using Theorem 4.5.3, we write

$$\begin{aligned} 1 - P(A_1 \cup \dots \cup A_n) &= P(A_1^c \cap A_2^c \cap \dots \cap A_n^c) \\ &= P(A_1^c)P(A_2^c) \cdots P(A_n^c) \\ &= [1 - P(A_1)][1 - P(A_2)] \cdots [1 - P(A_n)], \end{aligned}$$

as was to be shown. □

■ **EXAMPLE 4.16**

Consider a sequence of independent experiments such that in each of them an event A (usually labeled “success”) may or not occur. Let $P(A) = p$ for every experiment, and let A_k be the event “success at the k th trial.” Then $A_1 \cup \cdots \cup A_n$ is the event “at least one success in n trials,” and consequently

$$P(\text{at least one success in } n \text{ trials}) = 1 - (1 - p)^n.$$

At the end of this section, we will prove the theorem complementing Theorem 2.6.2.

Theorem 4.5.5 (Second Borel–Cantelli Lemma) *If $A_1, A_2, \dots$ is a sequence of independent events such that $\sum_{n=1}^{\infty} P(A_n) = \infty$, then*

$$P(\limsup A_n) = 1.$$

Proof: We have $\limsup A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \{\text{infinitely many } A_i \text{'s will occur}\}$. By De Morgan’s law, $[\limsup A_n]^c = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k^c$. Since the intersections $\bigcap_{k=n}^{\infty} A_k^c$ increase with n , we have

$$P([\limsup A_n]^c) = \lim_{n \rightarrow \infty} P\left(\bigcap_{k=n}^{\infty} A_k^c\right) = \lim_{n \rightarrow \infty} \lim_{N \rightarrow \infty} P\left(\bigcap_{k=n}^N A_k^c\right).$$

If A_k are independent, so are A_k^c . Then, using the inequality $1 - x \leq e^{-x}$, we obtain

$$\begin{aligned} \lim_{N \rightarrow \infty} P\left(\bigcap_{k=n}^N A_k^c\right) &= \lim_{N \rightarrow \infty} [1 - P(A_n)] \cdots [1 - P(A_N)] \\ &\leq \lim_{N \rightarrow \infty} e^{-[P(A_n) + \cdots + P(A_N)]} = 0 \end{aligned}$$

in view of the assumption that the series $\sum P(A_n)$ diverges. Consequently,

$$P([\limsup A_n]^c) = 0,$$

as was to be shown. □

PROBLEMS

- 4.5.1** Label the following statements as true or false: **(i)** The target is to be hit at least once. In three independent shots at the target (instead of one shot), you triple the chances of attaining the goal (assume each shot has the same positive chance of hitting the target). **(ii)** If A and B are independent, then $P(A^c|B^c) = 1 - P(A)$. **(iii)** If A and B are independent, then they must be disjoint. **(iv)** If A and B are independent, $P(A) = P(B)$, and $P(A \cup B) = 1/2$, then $P(A) > 1/4$.
- 4.5.2** Let A and B be independent events. Find $P(A \cap B)$ if: **(i)** $P(A \cap B^c) = 1/3$ and $P(A^c \cap B) = 1/6$. **(ii)** $P(A) = kP(B)$, and either A or B must occur.
- 4.5.3** A die is tossed three times, with outcomes X_1, X_2 , and X_3 . Assuming that all 216 possible outcomes (x_1, x_2, x_3) are equally likely, find following probabilities: **(i)** $P(X_1 > X_2 = X_3)$. **(ii)** $P(X_1 < X_2 < X_3)$. **(iii)** $P[\max(X_1, X_2, X_3) = 4]$. **(iv)** $P[\min(X_1, X_2, X_3) = 3]$.

- 4.5.4** Events A and B are independent, A and C are mutually exclusive, and B and C are independent. Find $P(A \cup B \cup C)$ if $P(A) = 0.5$, $P(B) = 0.25$, and $P(C) = 0.125$.
- 4.5.5** If disjoint events A and B have positive probabilities, check independence of events in the following pairs: $\emptyset$ and A , A and B , A and S , A and $A \cap B$, $\emptyset$ and A^c .
- 4.5.6** Let X be the number on the chip randomly selected from a box containing 12 chips, labeled 1 through 12. Check pairwise independence of events A , B , and C , defined as X is even, $X \geq 7$, and $X < 4$, respectively.
- 4.5.7** The probability that a certain event A occurs at least once in three independent trials exceeds the probability that A occurs twice in two independent trials. Find the possible values of $P(A)$.
- 4.5.8** Suppose that a point is picked at random from the unit square $0 \leq x \leq 1, 0 \leq y \leq 1$. Let A be the event that it falls in the triangle bounded by the lines $y = 0$, $x = 1$, and $x = y$, and let B be the event that it falls into the rectangle with vertices $(0, 0)$, $(1, 0)$, $(1, 1/2)$, and $(0, 1/2)$. Find all statements that are true: (i) $P(A) = P(B)$. (ii) $P(A \cap B) = 3/8$. (iii) $P(A \cup B) = P(A \cap B)$. (iv) A and B are independent. (v) $P(A|B) = 1/8$.
- 4.5.9** Events A and B are such that $3P(A) = P(B) = p$, where $0 < p < 1$. Find the correct answers in parts (i) and (ii). (i) The relation $P(B|A) = 3P(A|B)$ is (a) True. (b) True only if A and B are disjoint. (c) True only if A and B are independent. (d) False. (ii) The relation $P(A \cap B^c) \leq \min(p/3, 1 - p)$ is (a) True. (b) False.
- 4.5.10** A coin is tossed six times. Find the probability that the number of heads in the first three trials is the same as the number of heads in the last three trials.
- 4.5.11** Two people take turns rolling a die. Peter rolls first, then Paul, then Peter again, and so on. The winner is the first to roll a six. What is the probability that Peter wins?
- 4.5.12** A coin with probability p of turning up heads is tossed until it comes up tails. Let X be the number of tosses required. You bet that X will be odd, and your opponent bets that X will be even. For what p is the bet advantageous to you? Is there a p such that the bet is fair?
- 4.5.13** Find the probability that in repeated tossing of a pair of dice, a sum of 7 will occur before a sum of 8.
- 4.5.14** Three people, A, B, and C, take turns rolling a die. The first one to roll 5 or 6 wins, and the game is ended. Find the probability that A will win.
- 4.5.15** Consider a die in which the probability of a face is proportional to the number of dots on this face. What is the probability that in six independent throws of this die each face appears exactly once?
- 4.5.16** A machine has three independent components, two that fail with probability p and one that fails with probability 0.5. The machine operates as long as at least two parts work. Find the probability that the machine will fail.
- 4.5.17** The French mathematician Jean D'Alembert claimed that in tossing a coin twice, we have only three possible outcomes: "two heads," "one head," and "no heads." This is a legitimate sample space, of course. However, D'Alembert also claimed that each outcome in this space has the same probability $1/3$. (i) Is it possible to have a coin biased in such a way so as to make D'Alembert's claim true? (ii) Is it possible to make two coins with different probabilities of heads to confirm D'Alembert's claim?

4.5.18 Is it possible to bias a die in such a way that in tossing the die twice, each sum $2, 3, \dots, 12$ has the same probability?

4.6 EXCHANGEABILITY; CONDITIONAL INDEPENDENCE

We will now introduce an important type of dependence of events.

Definition 4.6.1 Let A, B , and H be three events, and let $P(H) > 0$. We say that events A and B are *conditionally independent given H* , if

$$P(A \cap B|H) = P(A|H)P(B|H). \quad \square$$

Definition 4.6.2 Let $\mathcal{H} = \{H_1, H_2, \dots\}$ be a positive partition (finite or countably infinite). We say that events A and B are *conditionally independent given $\mathcal{H}$* , if A and B are conditionally independent given any set H_i in partition $\mathcal{H}$. $\square$

■ EXAMPLE 4.17

In order to understand why events that are conditionally independent with respect to every set of the partition may be dependent, consider the following simple situation: Let the partition consist of two events, H and H^c , with $P(H) = 0.6$. Suppose that if H occurs, then both A and B are likely, say $P(A|H) = 0.8$, $P(B|H) = 0.9$, and conditional independence requires that $P(A \cap B|H) = 0.8 \times 0.9 = 0.72$. On the other hand, if H^c occurs, then both A and B are rather unlikely, say $P(A|H^c) = P(B|H^c) = 0.1$. Again, conditional independence requires that $P(A \cap B|H^c) = (0.1)^2 = 0.01$.

It is easy to check that the events A and B are dependent. Indeed, $P(A) = P(A|H)P(H) + P(A|H^c)P(H^c) = 0.8 \times 0.6 + 0.1 \times 0.4 = 0.52$, $P(B) = P(B|H)P(H) + P(B|H^c)P(H^c) = 0.9 \times 0.6 + 0.1 \times 0.4 = 0.58$. On the other hand, $P(A \cap B) = P(A \cap B|H)P(H) + P(A \cap B|H^c)P(H^c) = 0.72 \times 0.6 + 0.01 \times 0.4 = 0.436$, which is not equal to $P(A)P(B) = 0.3016$.

What is happening here is that the occurrence or nonoccurrence of one of the events allows us to make rather reliable predictions about the other event. In the case under considerations, the events A and B are positively correlated: If one occurs, the other is likely to occur also, and vice versa. By Definition 4.5.2 we have

$$r(A, B) = \frac{0.436 - 0.3016}{\sqrt{0.52 \times 0.48 \times 0.58 \times 0.42}} = 0.545.$$

The nature of dependence that we have here might better be visualized by a real-life situation. Imagine some animals, such as birds, that raise their young every year. If the conditions in a given year are hard (a draught, severe winter, etc.), often all the young die, and typically very few survive. If the conditions are favorable, the number of young that make it to next year is typically higher. The fates of different families *in any given year* are independent, in the sense that survival of offspring in one family does not depend on the survival of offspring in the other. If A and B are events occurring in two different families, and H_i 's are events describing possible types of conditions in any given year, then A and B are conditionally independent with respect to any given events H_i .

Unconditionally, however, events A and B are dependent, the dependence arising from the fact that all families are subject to the same conditions. Thus, if one family

fares badly, then we can expect that others will also, simply because the conditions are likely to be hard for them all.

The definition of conditional independence given an event, and therefore also definition of conditional independence given a partition, extends naturally to the case of n events. The extension consists of first defining the conditional independence of events $A_1, A_2, \dots, A_n$ given an event H_i . We will not give the definition here. It is a repetition of Definition 4.5.3, the only difference being that the unconditional probabilities in formula (4.18) are replaced by conditional probabilities given H_i . Definition 4.6.2 remains unchanged.

Now let $A_1, A_2, \dots, A_n$ be a set of events, and let $N = \{1, 2, \dots, n\}$ be the set of their indices. We introduce the following definition:

Definition 4.6.3 The events $A_1, A_2, \dots, A_n$ are called *exchangeable* if for any subset K of $\{1, 2, \dots, n\}$ the probability

$$P\left(\bigcap_{i \in K} A_i \cap \bigcap_{j \notin K} A_j^c\right)$$

of joint occurrence of all events with indices in K and nonoccurrence of all events with indices not in K depends only on the size of the set K . $\square$

This means that for exchangeable events, the probability of occurrence of exactly m of the events does not depend on which events are to occur and which are not. For instance, in the case of three events A, B , and C , their exchangeability means that $P(A \cap B^c \cap C^c) = P(A^c \cap B \cap C^c) = P(A^c \cap B^c \cap C)$ and also $P(A \cap B \cap C^c) = P(A \cap B^c \cap C) = P(A^c \cap B \cap C)$.

We have the following theorem:

Theorem 4.6.1 If the events $A_1, A_2, \dots, A_n$ are conditionally independent given a partition $\mathcal{H} = \{H_1, H_2, \dots\}$, and $P(A_i|H_k) = P(A_j|H_k)$ for all i, j, k , then they are exchangeable.

Proof: Let $P(A_i|H_k) = w_k$ (by assumption, this probability does not depend on i). The probability that, say, the first r events will occur and the remaining ones will not (by the law of total probability and conditional independence) is

$$\begin{aligned} & P(A_1 \cap A_2 \cap \dots \cap A_r \cap A_{r+1}^c \cap \dots \cap A_n^c) \\ &= \sum_k P(A_1 \cap A_2 \cap \dots \cap A_r \cap A_{r+1}^c \cap \dots \cap A_n^c | H_k) P(H_k) \\ &= \sum_k w_k^r (1 - w_k)^{n-r} P(H_k). \end{aligned} \tag{4.19}$$

Clearly, (4.19) is also the probability of occurrence of *any* r among the events $A_1, \dots, A_n$ and nonoccurrence of the remaining ones. $\square$

As can be expected, in case of conditional independence of A' and A'' with respect to the partition $\mathcal{H}$, Theorem 4.4.2 on updating the evidence will take a simpler form. We have

Theorem 4.6.2 Let $\mathcal{H} = \{H_1, H_2, \dots\}$ be a partition and let A', A'' be two events conditionally independent with respect to $\mathcal{H}$. If $P(A' \cap A'') > 0$, then for every event H_k in partition $\mathcal{H}$ we

have

$$\begin{aligned}
 P(H_k|A' \cap A'') &= \frac{P(A'|H_k)P(A''|H_k)P(H_k)}{\sum_j P(A'|H_j)P(A''|H_j)P(H_j)} \\
 &= \frac{P(A''|H_k)P(H_k|A')}{\sum_j P(A''|H_j)P(H_j|A')}.
 \end{aligned} \tag{4.20}$$

Proof: The middle part shows how the conditional probability of H_k is computed if the “evidence” $A' \cap A''$ is taken jointly. The last part shows how the updating is done successively, if first one updates probabilities using A' and then one uses A'' for further modification of posterior probabilities.

For the first equation, observe that the middle part of (4.20) is just the middle part of (4.14) in the case of conditional independence.

To prove the second equation, observe that the factor $P(A''|H_k \cap A')$ in the right-hand side of (4.14) can be written as

$$\begin{aligned}
 P(A''|H_k \cap A') &= \frac{P(A' \cap A'' \cap H_k)}{P(A' \cap H_k)} = \frac{P(A' \cap A''|H_k)P(H_k)}{P(A'|H_k)P(H_k)} \\
 &= \frac{P(A'|H_k)P(A''|H_k)}{P(A'|H_k)} = P(A''|H_k).
 \end{aligned}$$

□

PROBLEMS

- 4.6.1** A subset S of size t , $1 \leq t \leq N$, is selected at random from the set $\{1, \dots, N\}$ and event A_i , $i = 1, \dots, n$, is defined as “Element i was among the elements selected.” If $S = \{i_1, \dots, i_t\}$ is chosen, we say that events $A_{i_1}, \dots, A_{i_t}$ occur, while the remaining events do not. **(i)** Show that events $A_1, \dots, A_N$ are exchangeable. **(ii)** Find the correlation coefficient $r = r(A_i, A_j)$, $i \neq j$ between two events. Give an intuitive explanation why $r < 0$.
- 4.6.2** Generalizing the scheme of Problem 4.6.1, let p_t be the probability of choosing size t for subset $S_t \subset \{1, \dots, N\}$, $1 \leq t \leq N$. After choosing t , subset S_t is selected at random, and all events with indices in S_t occur while other events do not. **(i)** Argue that events $A_1, \dots, A_N$ are exchangeable. **(ii)** Find probabilities $P(A_1^c)$ and $P(A_1 \cap A_2)$.

4.7 MARKOV CHAINS*

With the material introduced so far it is already possible to develop an extensive and powerful theory with numerous practical applications and cognitive consequences, namely the theory of Markov chains. While the theory of statistical processes and Markov chains, in particular, is beyond the scope of this book, it has numerous practical applications and therefore we will introduce the basic concepts here.

Generally, the term *chain* will denote a sequence of events, typically dependent one on another in some way. It will be convenient to use the terminology referring to time: we can think of events as occurring one after another so that the event whose occurrence is actually observed is the “present” event, while the events following it belong to the “future” and the remaining ones to the “past.” This way we obtain a description of a process of dynamic changes (of something that we generally call a *system*). Quite naturally the main problem

will be to develop tools for making some inference (prediction, etc.) of the future on the basis of the knowledge of the past.

Specifically, consider a system that evolves in a random manner. The observations are taken at some fixed times $t_0 < t_1 < t_2 < \dots$, so in effect we record a sequence of states at which we find the system at the times of the observations. Without loss of generality, we let $t_n = n$ so that the observations are taken every unit of time starting at $t_0 = 0$.

The notions of “system” and “state” in the description above are left unspecified in order to have flexibility in applying the theory to various situations. Before proceeding any further, let us consider some examples that will later guide our intuition.

■ EXAMPLE 4.18

Imagine a gambler who plays a sequence of games. In each game, she may win or lose some amount of money. Let w_n denote her net winnings in n th game (loss, if w_n is negative). If we are interested only in the financial aspects of the situation, we can regard the gambler as a “system,” with the state of the system identified with the gambler’s fortune. Thus, the state in this case is a number. Letting $s(n)$ be the state at the time immediately after n th game, and letting $s(0)$ be the initial fortune, we have $s(n) = s(0) + w_1 + w_2 + \dots + w_n$.

This scheme has to be made specific by adding appropriate assumptions. First, we need assumptions about the nature of the winnings w_k . In the simplest case, the gambler may play for the same fixed amount, say \$1, so that $s(n)$ either increases or decreases by 1 with each game, and the outcomes of consecutive games are independent. Another kind of assumption is that of “boundary” conditions, which specify what happens when the gambler becomes ruined, that is, when $s(n)$ reaches the value 0. For example, the game may end, with $s(n)$ remaining 0 forever, or it may continue from the state $s(n) = 1$ if the gambler borrows a dollar to continue playing, and so forth.

Such an evolving scheme is often called a “random walk” because one can interpret the state $s(n)$ as a position of a pawn that moves randomly along the x -axis, being shifted at the n th move by the amount w_n .

The interesting question is: What is the probability that after the n th move at time t , the distance $s(n) - s(0)$ from the initial location will satisfy the inequality $a < s(n) - s(0) < b$? We will return to this problem in later chapters (especially Chapter 9) as we develop tools for analyzing the properties of random sums of the form $w_1 + \dots + w_n$, representing total displacement after n hits.

■ EXAMPLE 4.19

Consider now the evolution of an epidemic. Imagine a group of subjects, of whom some may be infected with the disease and therefore capable of spreading it to others (even though they may not yet be aware of being infectious), and some other persons may be susceptible. The spread of the disease is generated by the process of contact whereby proximity between an infective and a susceptible subject can lead to an infection. The state of the system can be described by a pair of numbers $s(n) = [x(n), y(n)]$, where $x(n)$ is the number of infectives at time $t = n$, and $y(n)$ is the number of susceptibles at time $t = n$.

By taking a sufficiently small time between observations, one can assume that the only possible changes from $[x(n), y(n)]$ to $[x(n+1), y(n+1)]$ are as follows:

1. $x(n+1) = x(n) + k, y(n+1) = y(n) - k$ (infection of a group of k susceptibles, $k = 1, 2, \dots$).

2. $x(n+1) = x(n) - 1, y(n+1) = y(n)$ (“removal” of an infective, which may correspond to death, isolation in the hospital, recovery with immunity, etc.).
3. “Status quo,” that is, $x(n+1) = x(n), y(n+1) = y(n)$.

The probabilities of these three types of transitions depend on the nature of the contact process and the mode of transmission of the given disease (in particular, the process simplifies somewhat if we assume that k is always 1). By specifying these probabilities, one may obtain models of spread of various infectious diseases.

Examples 4.18 and 4.19 show that the notions of the *system* and *state* are rather flexible and can be interpreted in a number of ways. Basically, *system* here means any fragment of reality that we want to model and analyze, while *state* is the information about the system that is relevant for the purpose of the study.

In this chapter, we will regard time as discrete. In general theory, one also considers systems evolving in continuous time, when changes can occur and be observed at any time. The theory designed to describe and analyze randomly evolving systems is called the *theory of stochastic processes*. While such a theory lies beyond the scope of the present book, we will occasionally introduce some elements of it.

It ought to be clear from the Examples 4.18 and 4.19 that a sample space that may be useful for describing the evolution of a system consists of all possible sequences of states at the observation times. Each such sequence represents a possible history of the process. Formally, let $\mathcal{E}$ be the set of all possible states of the system, with elements of $\mathcal{E}$ denoted by letter e , possibly with identifying subscripts or superscripts, such as e_1, e' . The sample space $\mathcal{S}_N$, which will represent the history of the system at times $t = 0, 1, \dots, N$, will consist of all possible sequences $s_N = [s(0), \dots, s(N)]$, where $s(i)$ is the element of $\mathcal{E}$ describing the state of the system at time $t = i$.

We will assume that the set $\mathcal{E}$ is either finite or countably infinite. We may then label the states by integers so that $\mathcal{E} = \{e_1, e_2, \dots, e_n\}$ or $\mathcal{E} = \{e_1, e_2, \dots\}$. Under this assumption each of the sample spaces $\mathcal{S}_N$ is also either finite or countably infinite, and we will define the probabilities on individual sample points of $\mathcal{S}_N$ by explicit formulas. At the same time, however, we will consider certain limiting passages, with the lengths of the sequences of states increasing to infinity. The space of all infinite sequences of states in $\mathcal{E}$ is uncountable whether or not $\mathcal{E}$ is finite. The details of the construction of probability of such spaces (although sketched in Chapter 2) lie beyond the scope of this book. However, we will use simple limiting passages with the lengths of sequences increasing to infinity, basically treating each space $\mathcal{S}_N$ as a space of all beginnings of length N of infinite sequences.

The main types of events that we will consider are of the form “state e at time $t = n$ ”. In Chapter 1, events were identified with subsets of the sample space. Accordingly, the event above can be defined as

$$\begin{aligned} \text{“state } e \text{ at time } t = n\text{”} &= \{s(n) = e\} \\ &= \text{set of all sequences } [s(0), \dots, s(N)] \\ &\quad \text{with } s(n) \text{ equal } e. \end{aligned}$$

Typically, the probability of the next state depends in some way on the preceding states. So by the chain formula (4.5), for all $n \leq N$, we have

$$\begin{aligned} P[s(0) = e_{i_0}, s(1) = e_{i_1}, \dots, s(n) = e_{i_n}] &= P[s(0) = e_{i_0}] \\ &\quad \times P[s(1) = e_{i_1} | s(0) = e_{i_0}] \times \dots \times P[s(n) = e_{i_n} | s(n-1) = e_{i_{n-1}}, \dots, s(0) = e_{i_0}]. \end{aligned} \tag{4.21}$$

The commas in formula (4.21) signify the operation of an intersection of events, thus replacing a rather clumsy notation $P[s(0) = e_{i_0} \cap s(1) = e_{i_1} \cap \dots \cap s(n) = e_{i_n}]$. For convenience, the events in the conditions are written in the order that starts from the most recent one. Formula (4.21) expresses the probability of a sequence of consecutive states through the probability of the “initial” state $s(0)$, and conditional probabilities involving all preceding states, and thus encompasses longer and longer fragments of the past. By the definition below, such conditional probabilities for Markov chains can be simplified by reaching only to the most recent state.

Definition 4.7.1 The evolution of a system is said to form a *Markov chain* if for every n and every vector $(e_{i_0}, e_{i_1}, \dots, e_{i_{n-1}}, e_{i_n})$ of states, the conditional probabilities satisfy the following relation, called the *Markov property*:

$$\begin{aligned} P[s(n) = e_{i_n} | s(n-1) = e_{i_{n-1}}, s(n-2) = e_{i_{n-2}}, \dots, s(0) = e_{i_0}] \\ = P[s(n) = e_{i_n} | s(n-1) = e_{i_{n-1}}]. \end{aligned} \quad (4.22)$$

□

We have therefore the next theorem.

Theorem 4.7.1 *If the sequence of transitions between states constitutes a Markov chain, then*

$$\begin{aligned} P[s(0) = e_{i_0}, \dots, s(n) = e_{i_n}] \\ = P[s(0) = e_{i_0}] \prod_{j=1}^n P[s(j) = e_{i_j} | s(j-1) = e_{i_{j-1}}]. \end{aligned} \quad (4.23)$$

Generally, the transition probability $P[s(j) = e_{i_j} | s(j-1) = e_{i_{j-1}}]$ from state $e_{i_{j-1}}$ to state e_{i_j} at time $t = j$ depends on j . If this is *not* the case, the situation can be greatly simplified, and this special case is the starting point of a theory. This theory is practically and cognitively useful, leads to rather deep mathematical results and fruitful generalizations, and is pleasantly elegant as well. Accordingly, we introduce the following definition:

Definition 4.7.2 A Markov chain will be said to have *stationary transition probabilities*, or be *time homogeneous*, if for all states e_i, e_j the probability

$$P[s(t) = e_j | s(t-1) = e_i]$$

does not depend on t .

□

From now on, *all Markov chains under consideration in this chapter will have stationary transition probabilities*. Writing $p_{ij} = P[s(t) = e_j | s(t-1) = e_i]$ and $r_i = P[s(0) = e_i]$, we may recast (4.23) as

$$P[s(0) = e_{i_0}, s(1) = e_{i_1}, \dots, s(n) = e_{i_n}] = r_{i_0} p_{i_0 i_1} \dots p_{i_{n-1} i_n}.$$

Clearly, $\{r_i, i = 1, 2, \dots\}$, called the *initial probability distribution*, and the set of probabilities p_{ij} , called the *transition probability matrix*, must satisfy the following conditions:

$$\sum_i r_i = 1 \quad (4.24)$$

and

$$\sum_j p_{ij} = 1 \quad \text{for every } i. \quad (4.25)$$

The last condition means simply that if the system is in some state e_i , it must pass to one of the states e_j (possibly remaining in the same state e_i) in the next step.

Since square matrices (finite or not) with nonnegative entries and row sums equal to 1 are called *stochastic matrices*, we see—in view of (4.25)—that every transition probability matrix of a Markov chain is a stochastic matrix.

■ EXAMPLE 4.20 Gambler's Ruin

Continuing Example 4.18 about the gambler, assume that each game is played for a unit stake, with the gambler's probability of winning a game being p . Moreover, assume that the games are independent. In such a case, the sequence $s(0), s(1), \dots$ of the gambler's fortunes after consecutive games is a Markov chain. Indeed, letting $P[s(n+1) = j | s(n) = i] = p_{ij}$, we have for all $k > 0$,

$$p_{k,k+1} = p, \quad p_{k,k-1} = 1 - p,$$

and $p_{kj} = 0$ for all j other than $k+1$ or $k-1$. These probabilities do not depend on the history $s(n-1), s(n-2), \dots$ preceding the n th game. The transition probabilities for $k=0$ depend on the assumption about what happens when the gambler becomes ruined. For instance, if upon reaching $k=0$ the game stops, we can put p_{0j} equal to 1 for $j=0$ and equal to 0 for all other j . If the process starts with the gambler having initial capital M , then r_j equals 1 for $j=M$ and equals 0 for all other j . It seems plausible (and in fact it is true) that if $p < 1/2$ (the games are unfavorable for the gambler), then regardless of the initial state, the state 0 (gambler's ruin) will sooner or later be reached. Less obvious, but still true, is that the same holds for the fair games ($p = 1/2$) no matter what the initial fortune is. An interesting question is to determine the probability of the gambler ever reaching 0 (i.e., of becoming eventually ruined) if she plays favorable games, that is, if $p > 1/2$. This probability depends on the initial state M , and may be shown to be $[(1-p)/p]^M$.

The following examples illustrate the problem of *absorption*.

■ EXAMPLE 4.21 Division of Stake in Gambler's Ruin

Two players, A and B, play a sequence of games, each game for a unit stake, until one of them loses all his or her money. Games are independent; in each of the probability of A winning is p (and the probability of A losing is $1-p$, which means that there are no ties). For some reason, the contest is interrupted (and is not going to be resumed) when A has x dollars and B has $M-x$ dollars (so that the total stake is M). The question is: How can the stake M be divided between the players in a fair way? One of the ways (arguably just) is to divide the stake in proportion to the chances of eventually winning the contest if it was to continue. Consequently, the problem lies in determining the probabilities of eventually winning the contest for each of the two players. We can interpret the situation in terms of the gambler's problem as follows: Let $i = 0, 1, \dots, M$ represent the state of the system, defined as the capital of A. Then the changes of states constitute a Markov chain, as in Example 4.20, the only difference being that now $p_{00} = p_{MM} = 1$ (i.e., one cannot leave states 0 and M). The objective is to determine $u(x)$, equal to the probability of A ultimately winning the contest if the starting state is x .

To see the technique, observe that for $0 < x < M$ the probabilities $u(x)$ must satisfy the equations

$$u(x) = pu(x+1) + (1-p)u(x-1), \quad (4.26)$$

with boundary conditions $u(0) = 0, u(M) = 1$. Actually, (4.26) is a special case of the total probability formula (4.9) for the event “ultimate winning by A” and the partition into the two events “next game won by A” and “next game lost by A.” This technique of using the partition obtained by considering the possible results of the next transition will be used again and again in the analysis of Markov chains. In the present case, a linear combination of solutions of (4.26) is also a solution of (4.26). Consequently, we need two linearly independent solutions to determine the constants in their linear combinations using the two boundary conditions.

Obviously, $u(x) = 1$ is always a solution of (4.26). To find another solution, let $u(x) = s^x$. This gives $s = ps^2 + q$, where we let $q = 1 - p$. The solutions of the quadratic equation are $[1 \pm (1 - 4pq)^{1/2}]/2p = [1 \pm |p - q|]/2p$, which equal 1 and q/p . Thus, if $p \neq q$ (hence, $p \neq 1/2$), a general solution of (4.26) is $u(x) = A + B(q/p)^x$. Using the boundary conditions, we obtain

$$u(x) = \frac{(q/p)^x - 1}{(q/p)^M - 1}.$$

If $p = q = 1/2$, the solutions $u(x) = (q/p)^x$ and $u(x) = 1$ coincide. In this case as another (linearly independent) solution, we can take $u(x) = x$, and the general solution of (4.26) will be $u(x) = A + Bx$. Using again the boundary conditions, we obtain $u(x) = x/M$ as the probability of winning in the case of fair games.

■ EXAMPLE 4.22

Let us now return to Example 4.19 of an epidemic. The state of the system is represented by a pair of integers (x, y) , with x being the number of infectives and y being the number of susceptibles. As was argued, by taking sufficiently small time intervals between transitions, we can assume that apart from remaining in the same state, the only possible transitions from (x, y) are to $(x - 1, y)$, representing removal (death, isolation, recovery with immunity, etc.) of one infective, and to $(x + 1, y - 1)$, representing infection of one susceptible. These transitions are depicted in Figure 4.2 as

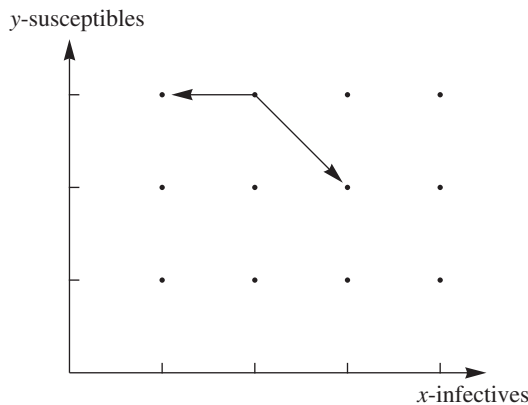


Figure 4.2 Transitions in model of epidemic.

a random walk over the integer lattice on the plane, with the two types of transitions being steps in the western and southeastern directions. We now have to define the transition probabilities for the two kinds of transitions above (removal and infection) so as to obtain a meaningful approximation to the real conditions of an epidemic. Here one could argue that it is reasonable to assume that the probability of the first transition (removal) is proportional to x , while the probability of the second transition (infection) is proportional to both x and y . Thus, the transition probabilities between states will be of the form

$$p[(x-1, y)|(x, y)] = ax, \quad p[(x+1, y-1)|(x, y)] = bxy$$

for suitably chosen a and b .

The nonlinear term (proportional to the product xy) is largely responsible for the fact that the analysis of an epidemic process is extremely difficult. From the point of view of applications, two characteristics of epidemics are of major interest: first, the time until the termination of the epidemic, and second, its total size, that is, the total number of persons who will become infected (and subsequently removed).

The first characteristic may be defined formally as

$$N = \text{first } n \text{ such that } x(n) = 0.$$

Here the event $N = k$ corresponds to $x(0) > 0, x(1) > 0, \dots, x(k-1) > 0$, and $x(k) = 0$. Graphically, N is the first time when the walk depicted in Figure 4.2 reaches the vertical axis.

The second characteristic cannot be expressed in terms of the variables introduced thus far. The handling of the case is an illustration of the point made repeatedly in Chapter 1 about the nonuniqueness of the choice of the sample space. One could simply enrich the state space by including another count that keeps track of the removals. Thus, one could consider the states as being described by three variables, $x(n)$, $y(n)$, and $z(n)$, where x and y are the numbers of infectives and susceptibles as before, while z is the number of removals. The two kinds of transitions considered so far are replaced by transitions from (x, y, z) to $(x-1, y, z+1)$, representing the removal of an infective, and from (x, y, z) to $(x+1, y-1, z)$, representing a new infection. Assuming that the epidemic starts with $z = 0$, the total size Z of epidemic is defined as $Z = z(N)$, where $N =$ the first n with $x(n) = 0$.

As mentioned above, finding the probabilities $P\{N = n\}$ and $P\{Z = k\}$ for given initial conditions of the Markov chain is extremely difficult, and we will not deal with this problem here.

We will now find the probability

$$p_{ij}^{(n)} = P\{s(t+n) = e_j | s(t) = e_i\} \quad (4.27)$$

of passing from e_i at time t to e_j at time $t+n$, called the n -step transition probability. Obviously, for Markov chains with stationary transition probabilities, the quantity (4.27) does not depend on t . We have the following theorem:

Theorem 4.7.2 *The n -step transition probabilities satisfy the relations*

$$p_{ij}^{(0)} = 1 \text{ or } 0, \text{ depending whether or not } j = i, \quad (4.28)$$

and for all $m, n \geq 0$

$$p_{ij}^{(m+n)} = \sum_k p_{ik}^{(m)} p_{kj}^{(n)}. \quad (4.29)$$

Proof: We proceed by induction. Formula (4.28) covers the case $m = n = 0$. In zero steps the system cannot change, so it must remain, with probability 1, in state e_i . We prove formula (4.29) for $n = 1$. If the system passes from e_i to e_j in $m + 1$ steps, then in step m it must be in some state e_k . The events “state e_k after m steps” form a partition, and the total probability formula (4.9) gives in this case $p_{ij}^{(m+1)} = \sum_k p_{ik}^{(m)} p_{kj}$, as was to be shown, since $p_{ij} = p_{ij}^{(1)}$ by definition. The extension to the case of arbitrary n is immediate. $\square$

In matrix notation, Theorem 4.7.2 states simply that $\mathbf{P}^{(m+n)} = \mathbf{P}^{m+n}$, with $\mathbf{P}^0 = \mathbf{I}$, the identity matrix.

One can expect that as the number of transitions increases, the system “forgets” the initial state; that is, the effect of the initial state on the probability of a state at time n gradually wears out. Formally, we can expect that the probabilities $p_{ij}^{(n)}$ converge, as n increases, to some limits independent of the starting state e_i . Let us consider the following example.

■ EXAMPLE 4.23

The weather on Markov Island is governed by the following laws. There are only three types of days: sunny, rainy, and cloudy, with the weather on a given day depending only on the weather on the preceding day. There are never two rainy days in row, and after a rainy day, a cloudy day is twice as likely as a sunny day. Fifty percent of days following a cloudy day are also cloudy, while 25% are rainy. Finally, after a sunny day, each type of weather is equally likely. How often, on average, is it cloudy on Markov Island?

SOLUTION. The fact that the tomorrow’s weather depends only on the weather today makes the process of the weather change a Markov process. Using the obvious notation R, S , and C for the three types of weather, we have $p_{RR} = 0$, $p_{RC} = 2/3$, $p_{RS} = 1/3$, $p_{CC} = 1/2$, and $p_{CR} = p_{CS} = 1/4$. Finally, $p_{SS} = p_{SC} = p_{SR} = 1/3$, so the transition matrix is

$$\mathbf{P} = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}. \quad (4.30)$$

Let us assume that the limits of the n -step transition probabilities exist and do not depend on the starting state. Thus, we have three limits, u_R, u_S , and u_C , where

$$u_R = \lim p_{RR}^{(n)} = \lim p_{CR}^{(n)} = \lim p_{SR}^{(n)},$$

and similarly for u_S and u_C . We can then write, using (4.29) for $n = 1$,

$$\begin{aligned} p_{RR}^{(m+1)} &= p_{RR}^{(m)} p_{RR} + p_{RC}^{(m)} p_{CR} + p_{RS}^{(m)} p_{SR}, \\ p_{CC}^{(m+1)} &= p_{CR}^{(m)} p_{RC} + p_{CC}^{(m)} p_{CC} + p_{CS}^{(m)} p_{SC}, \\ p_{SS}^{(m+1)} &= p_{SR}^{(m)} p_{RS} + p_{SC}^{(m)} p_{CS} + p_{SS}^{(m)} p_{SS}. \end{aligned} \quad (4.31)$$

Passing to the limit in (4.31), we obtain therefore a system of three linear equations:

$$\begin{aligned} u_R &= u_R p_{RR} + u_C p_{CR} + u_S p_{SR} \\ u_C &= u_R p_{RC} + u_C p_{CC} + u_S p_{SC} \\ u_S &= u_R p_{RS} + u_C p_{CS} + u_S p_{SS}. \end{aligned}$$

Using the probabilities from the matrix (4.30), we obtain

$$\begin{aligned} -u_R + \frac{u_C}{4} + \frac{u_S}{3} &= 0 \\ \frac{2u_R}{3} - \frac{u_C}{2} + \frac{u_S}{3} &= 0 \\ \frac{u_R}{3} + \frac{u_C}{4} - \frac{2u_S}{3} &= 0. \end{aligned} \tag{4.32}$$

This is a homogeneous system, so an additional equation is needed to determine the solution. We have here the identity

$$p_{RR}^{(n)} + p_{RC}^{(n)} + p_{RS}^{(n)} = 1, \tag{4.33}$$

and passing to the limit with n , we get

$$u_R + u_C + u_S = 1. \tag{4.34}$$

The solution of (4.32) and (4.34) is $u_R = 9/41$, $u_C = 20/41$, and $u_S = 12/41$, which means that almost 50% of days on Markov Island are cloudy.

The equations of this example are derived under the assumption that (1) the limits $p_{ij}^{(n)}$ exist and are independent of the starting state i , and (2) the number of states is finite. The latter assumption allowed us to pass to the limit in the sums in (4.31) and (4.33), obtaining (4.32) and (4.34). The first assumption here is true under some conditions. Basically, we have to exclude two obvious cases when the probabilities $p_{ij}^{(n)}$ either cannot converge, or if they do, the limits depend on the starting state i .

■ EXAMPLE 4.24

Consider a Markov chain with two states, e_1 and e_2 , such that $p_{12} = p_{21} = 1$ (so that $p_{11} = p_{22} = 0$). Clearly, the system must alternate between states 1 and 2 in a deterministic way. Consequently, $p_{11}^{(n)} = 0$ or 1, depending whether n is even or odd, and the sequence $p_{11}^{(n)}$ does not converge.

■ EXAMPLE 4.25

Consider again a Markov chain with two states e_1 and e_2 . Assume now that the transition probabilities are $p_{11} = p_{22} = 1$ (so that $p_{12} = p_{21} = 0$). In this case, the system must forever remain in the initial state, with $p_{11}^{(n)} = 1$ and $p_{21}^{(n)} = 0$ for all n . The limits $p_{ij}^{(n)}$ exist but depend on the initial state i .

These two examples, in essence, exhaust all possibilities that may prevent the existence of the limits $p_{ij}^{(n)}$ and their independence of the initial state i . One may summarize them as *periodicity* (in Example 4.24), and the *impossibility of reaching a certain state from some other state or states* (in Example 4.25).

PROBLEMS

4.7.1 Customers arrive at a service station (a taxi stand, a cable car lift at a skiing resort, etc.) and form a queue. At times $t = 1, 2, \dots$ the first m customers ($m \geq 1$) in the queue are served (if there are that many). Let $Y_1, Y_2, \dots$ denote the numbers of customers arriving during the time intervals between services, and let X_n denote the

number of customers in the system at the time immediately preceding the n th service. Argue that $X_{n+1} = (X_n - m)^+ + Y_n$, where $U^+ = \max(U, 0)$. Assuming that the events $\{Y_{n_1} = k_1\}, \dots, \{Y_{n_j} = k_j\}$ are independent for any $j, k_1, \dots, k_j$ and any distinct $n_1, \dots, n_j$, and that probabilities $P\{Y_n = k\} = p_k$, are independent of n , find $P(X_{n+1} = j | X_n = i)$ for all possible $i, j = 0, 1, 2, \dots$.

- 4.7.2** After some time spent in a bar, Peter starts to walk home. Suppose that the streets form a rectangular grid. Peter always walks to the nearest corner and then decides on the direction of the next segment of his walk (so that he never changes direction in the middle of the block). Define the “state of the system” in such a way as to accommodate conveniently the assumption that Peter never goes directly back to the corner he just left.
- 4.7.3 (i)** Modify Example 4.20 by assuming that in each game, the player may win with probability p , lose with probability q , or draw (so that his fortune does not change) with probability r , where $p + q + r = 1$. Find the transition probability matrix in this case. **(ii)** Modify Example 4.21 same way as in (i). Write the analogue to the equation (4.26) for the probability of ruin and solve it.
- 4.7.4** Suppose the results of an election in a certain city are found to depend only on the results of the last two elections. Specifically, letting R and D denote Republican and Democratic victories, the state before any election may be RR, RD, DR, DD, the letters signifying, respectively, the outcomes of the next-to-last and last elections. Generally, assume that a, b, c , and d are the probabilities of a Republican victory in each of the four states listed above. Find the transition probability matrix of the resulting Markov chain.
- 4.7.5** A college professor teaches a certain course year after year. He has three favorite questions, and he always uses one of them in the final exam. He never uses the same question twice in a row. If he uses question A in one year, then the next year, he tosses a coin to choose between question B and C. If he uses question B, he tosses a pair of coins, and chooses question A if both coins show tails. Finally, if he uses question C, then he tosses three coins and uses question A if all of them show heads. Find the transition probability matrix for the resulting Markov chain.
- 4.7.6 (Dog Fleas, or the Ehrenfest Model of Diffusion)** Consider two urns (or dogs), and N balls (or fleas), labeled $1, \dots, N$, allocated between the urns. At times $t = 1, 2, \dots$, a number 1 through N is chosen at random, and the ball with the selected number is moved to the other urn. Let $s(n)$ be the number of balls in the first urn at time n . Find the transition probability matrix for the Markov chain $s(n)$.
- 4.7.7** Consider a specific kind of part needed for the operation of a certain machine (e.g., the water pump of a car). When the part breaks down, it is replaced by a new one. The probability that a new part will last for exactly n days is $r_n, n = 1, 2, \dots$. Let the state of the system be defined as the age of the part currently in the machine. Find the transition probability matrix of the resulting chain.
- 4.7.8** Find all two-step transition probabilities for: **(i)** The Markov chain described in Problem 4.7.4. **(ii)** The dog flea model of Problem 4.7.6.
- 4.7.9** Assume that a man’s occupation can be classified as professional, skilled laborer, or unskilled laborer. Assume that of the sons of professional men, a percent are professional, the rest being equally likely to be skilled laborers or unskilled laborers. In the case of sons of skilled laborers, b percent are skilled laborers, the rest being equally likely to be professional men or unskilled laborers. Finally, in the case of

unskilled laborers, c percent of the sons are unskilled laborers, the rest again divided evenly between the other two categories. Assume that every man has one son. Form a Markov chain by following a given family through several generations. Set up the matrix of transition probabilities, and find: **(i)** The probability that the grandson of an unskilled laborer is a professional man. **(ii)** The probability that a grandson of a professional man is an unskilled laborer.

- 4.7.10** In Problem 4.7.9, it was assumed that every man had one son. Assume now that the probability that a man has a son is r . Define a Markov chain with four states, where the first three states are as in Problem 4.7.9, and the fourth state is entered when a man has no son. This state cannot be left (it corresponds to a male line of the family dying out). Find the probability that an unskilled laborer has a grandson who is a professional man.